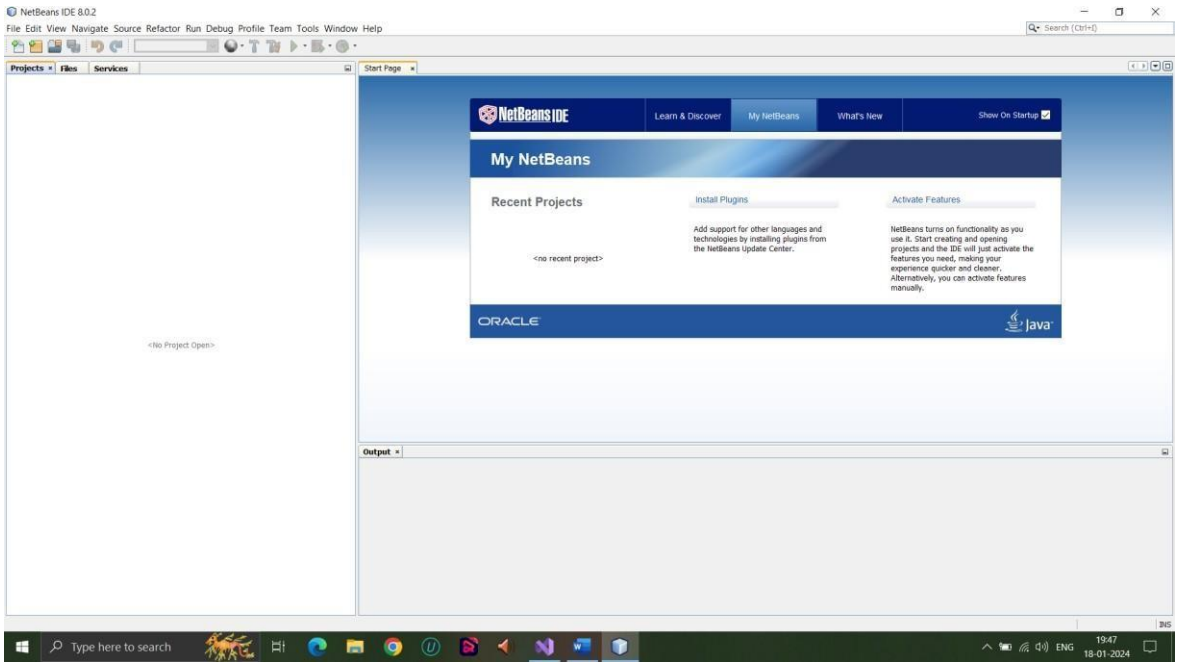
Practical-1

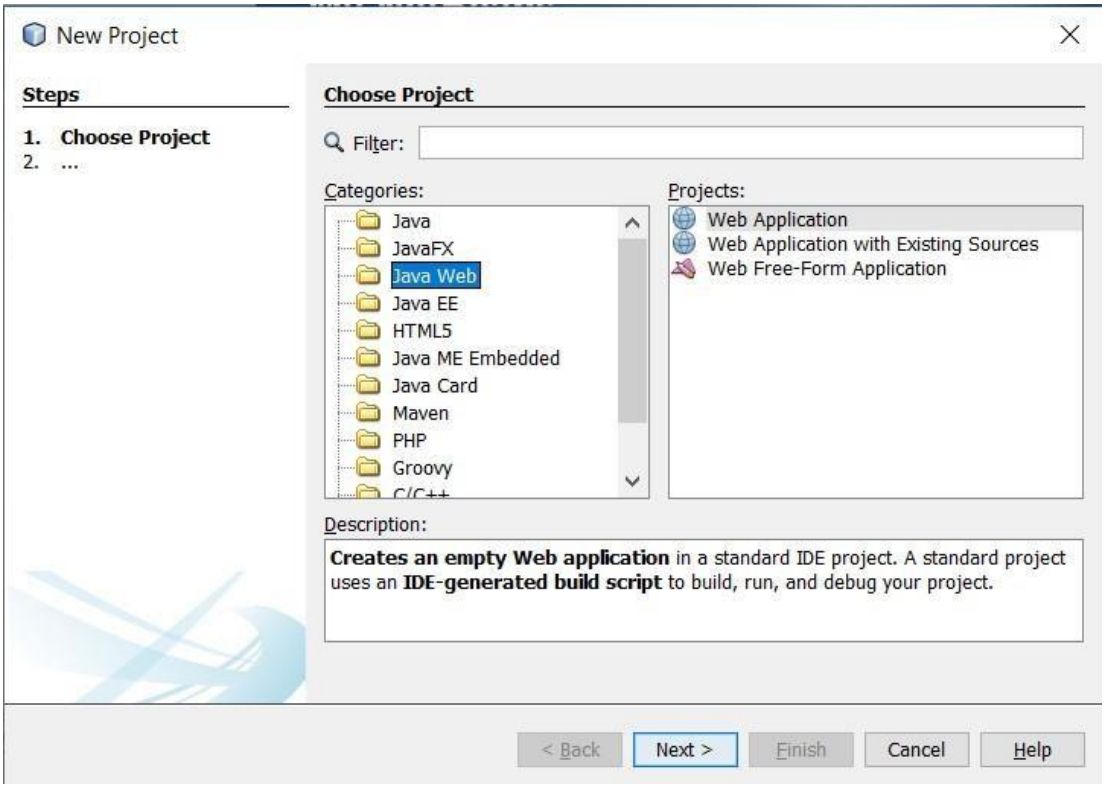
Create a Simple SOAP service.

Steps:

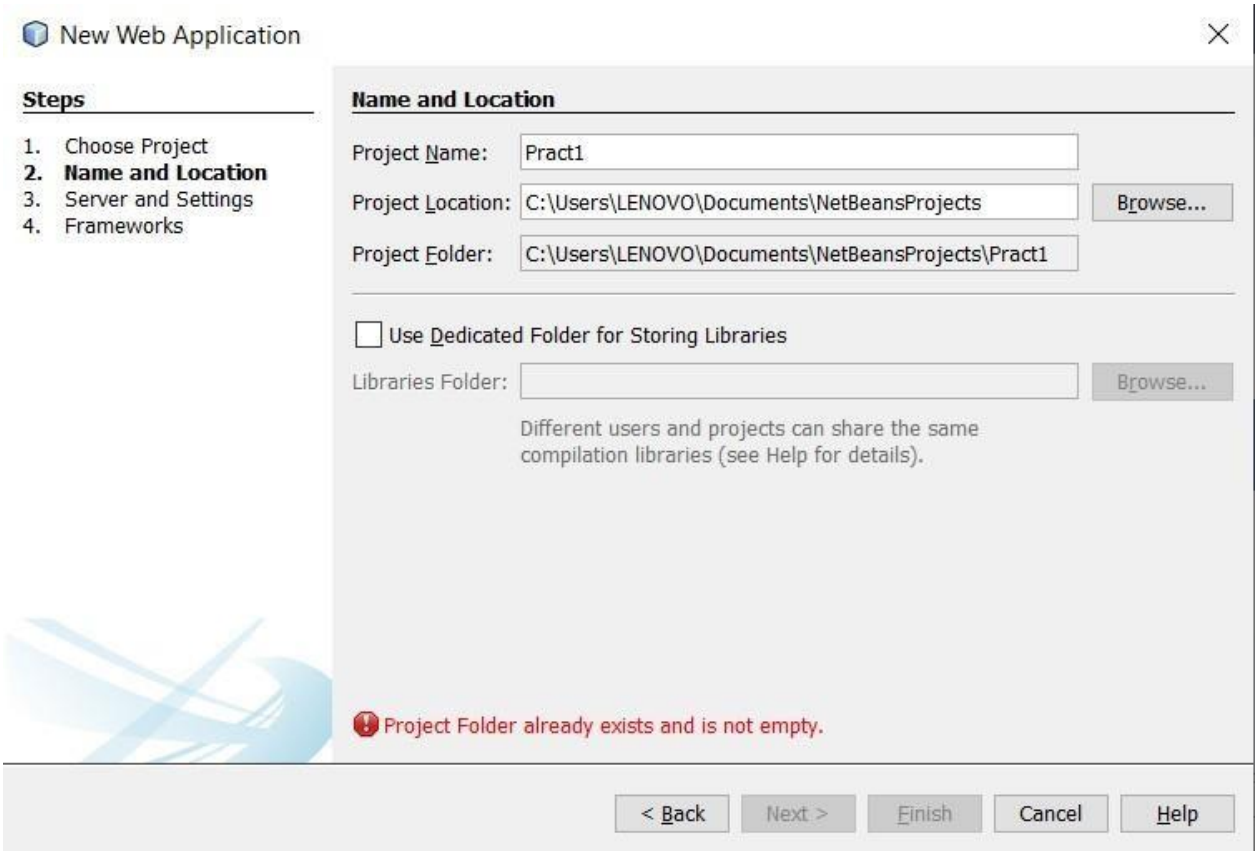
1] Open the NetBeans , and you will get the following screen. Close the start page.



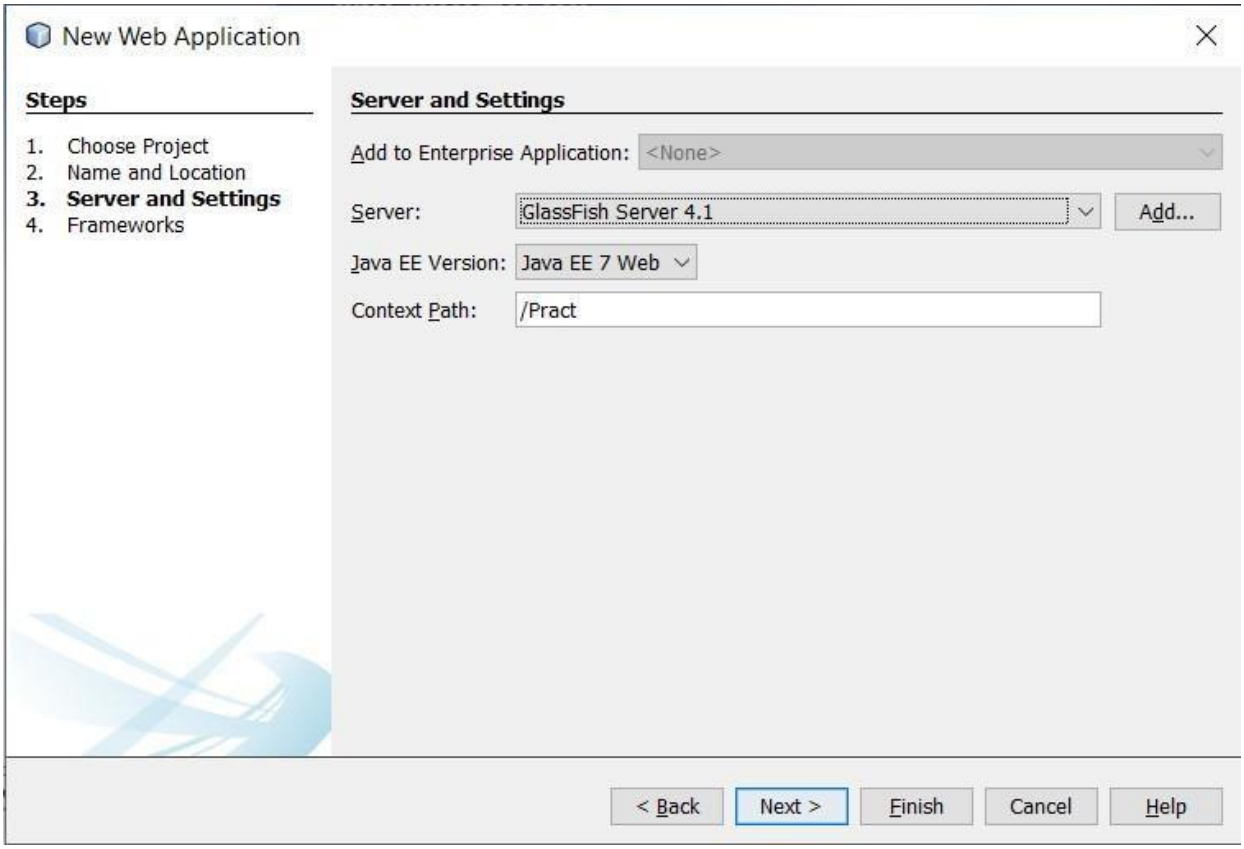
2] Now click on the file tab and click on new project you will get the following screen :



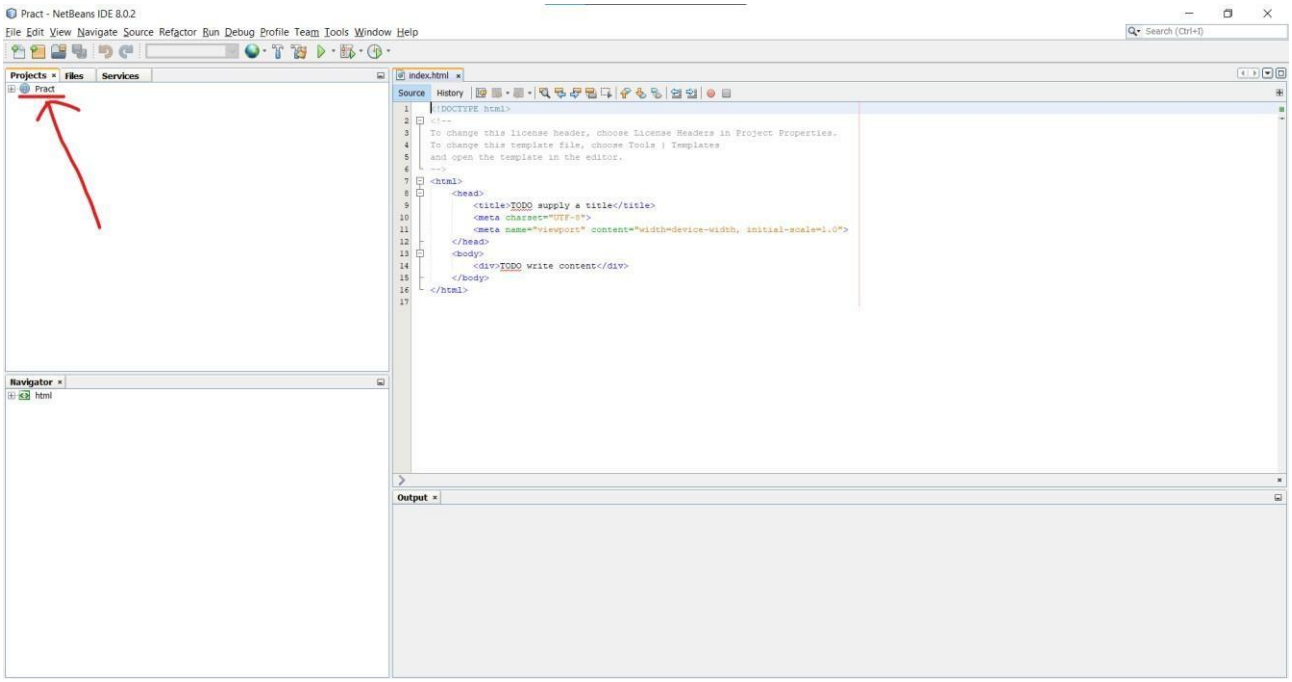
3] In Categories select Java Web and in Projects , Select Web Application .After selecting click onnext . You will get the following window:



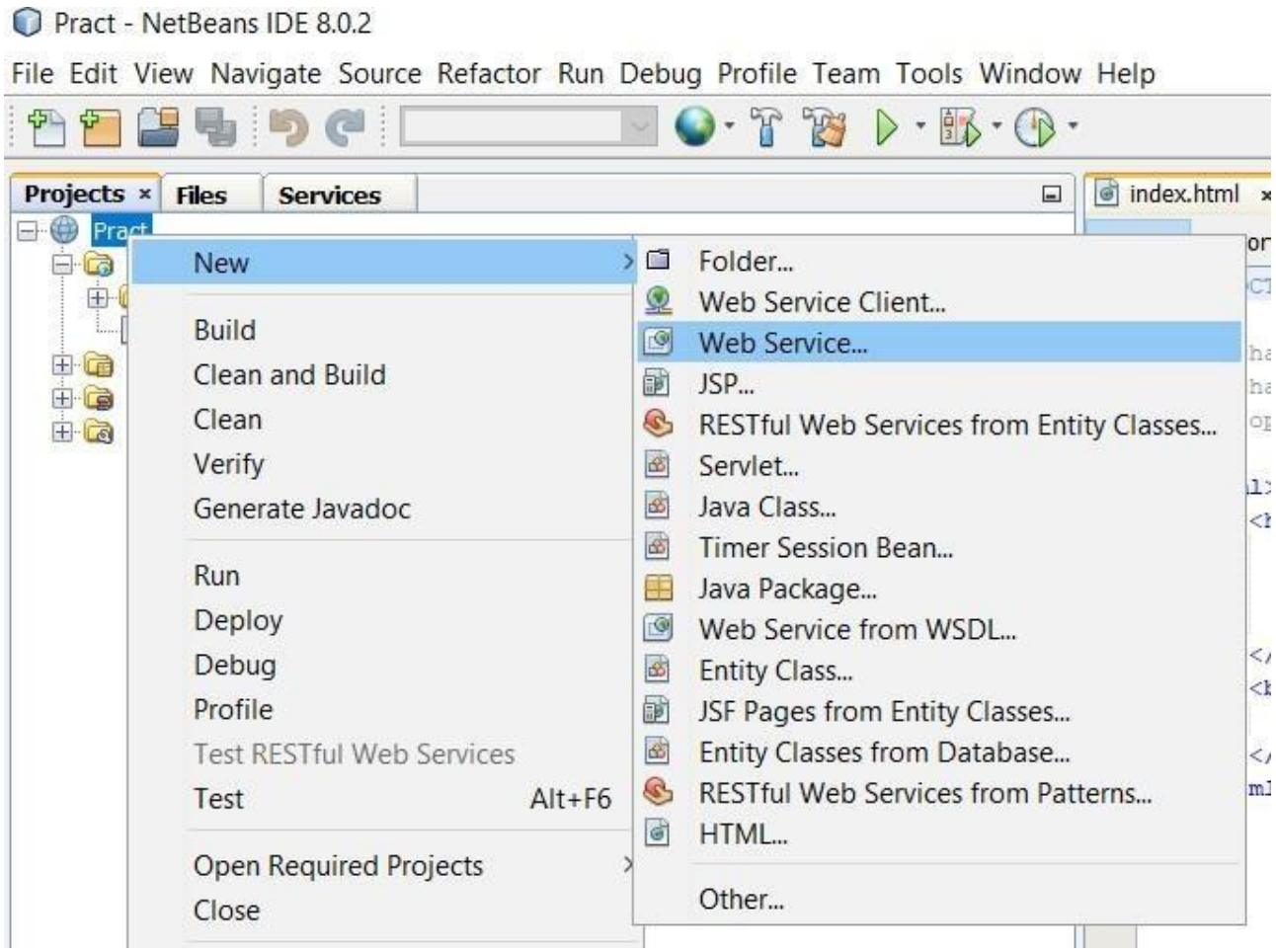
4] Now Give name to the Project Name: , and click on next . you will get the following windowthen . again click on finish:



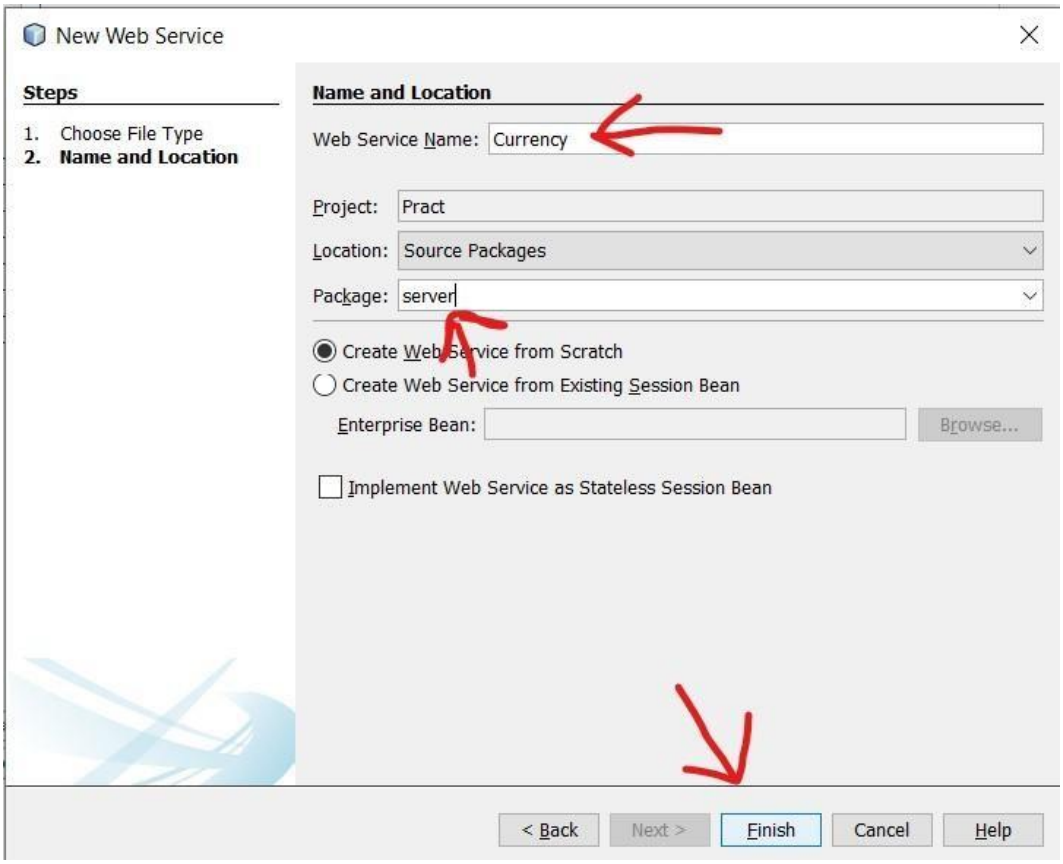
5] You will get the following screen now carefully see in projects section your recently created project appears double click to expand it:



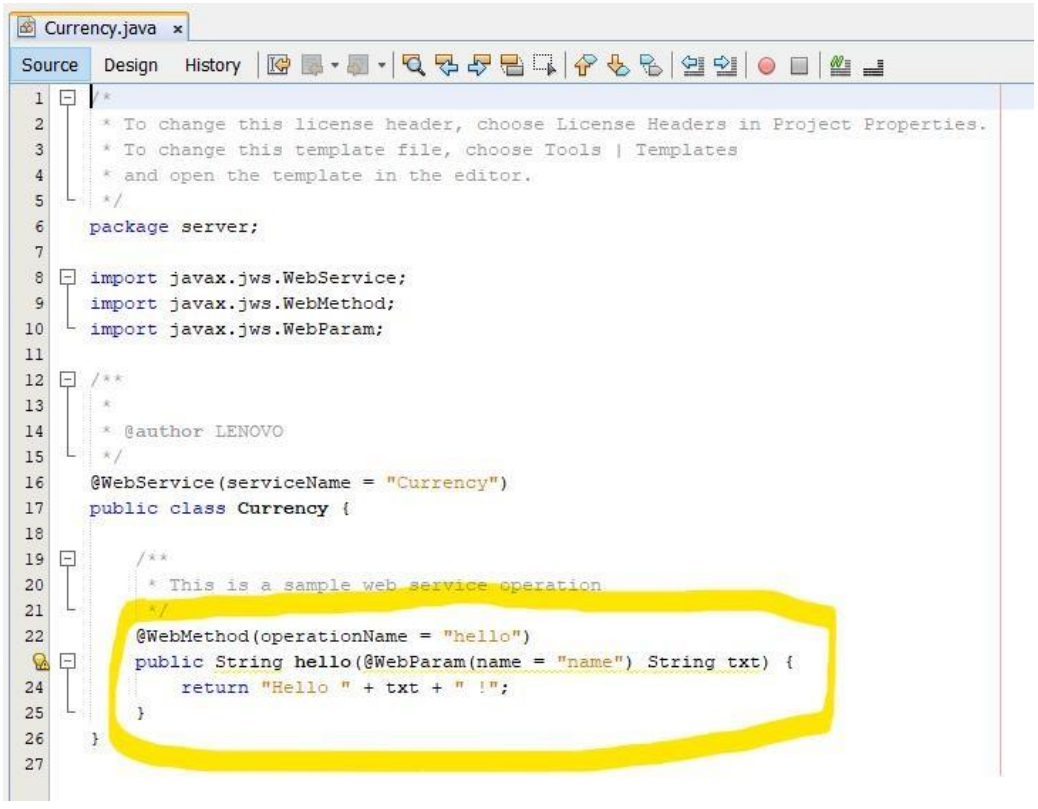
6] Now Right click on the project and select new and then select Web Service ,As shown Below:



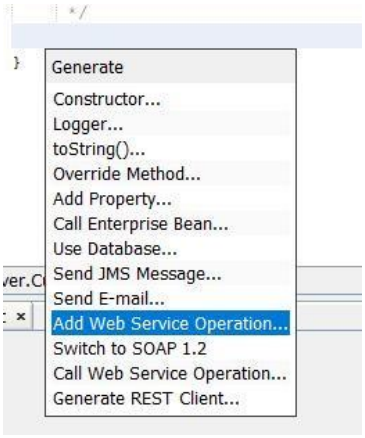
7] after clicking Web Service following window should appear , now give name to web serviceand give package as “server”. As shown in the image:



After clicking finish you should get the following window erase the mentioned code :

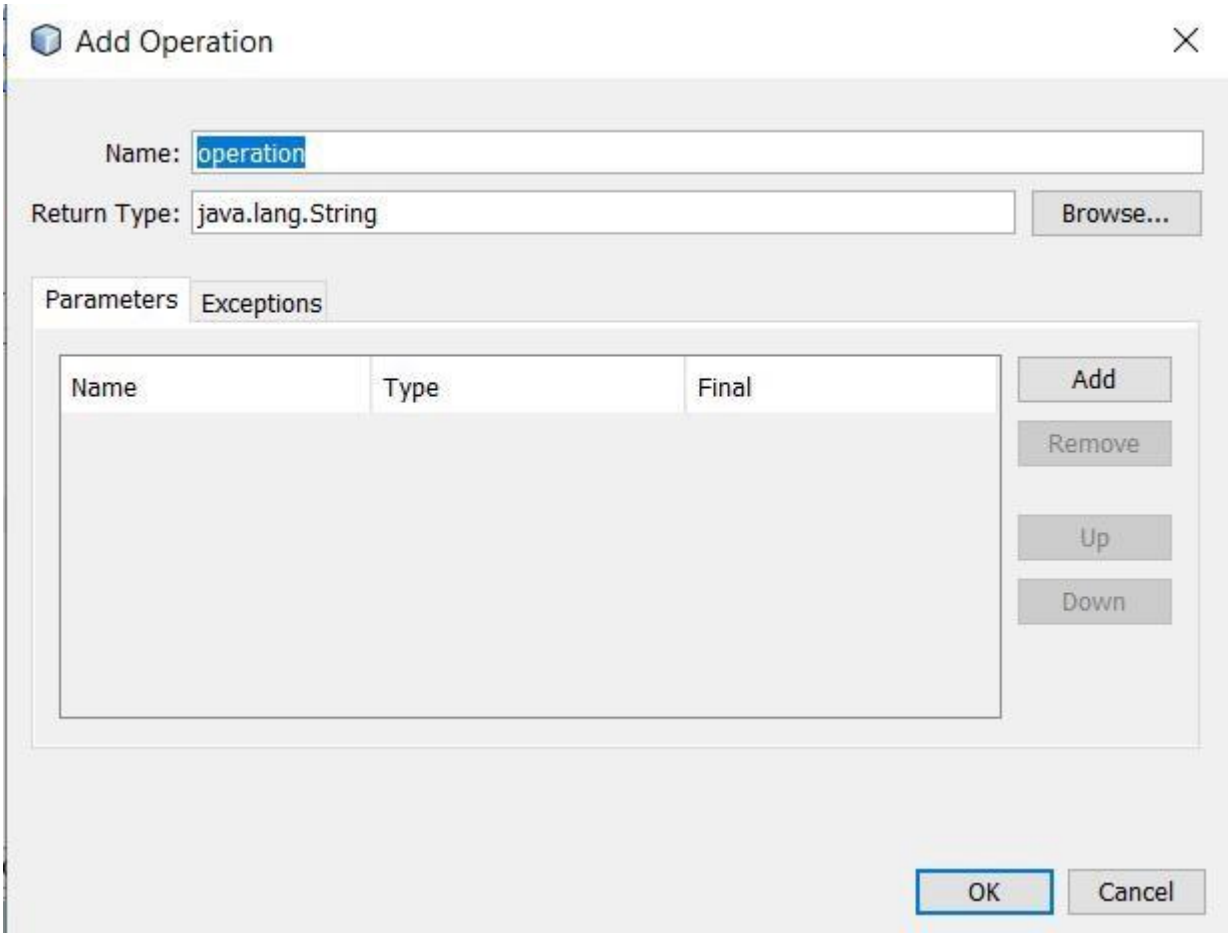


8] Now right click any where and click on insert code and select Add Web Service Operation :

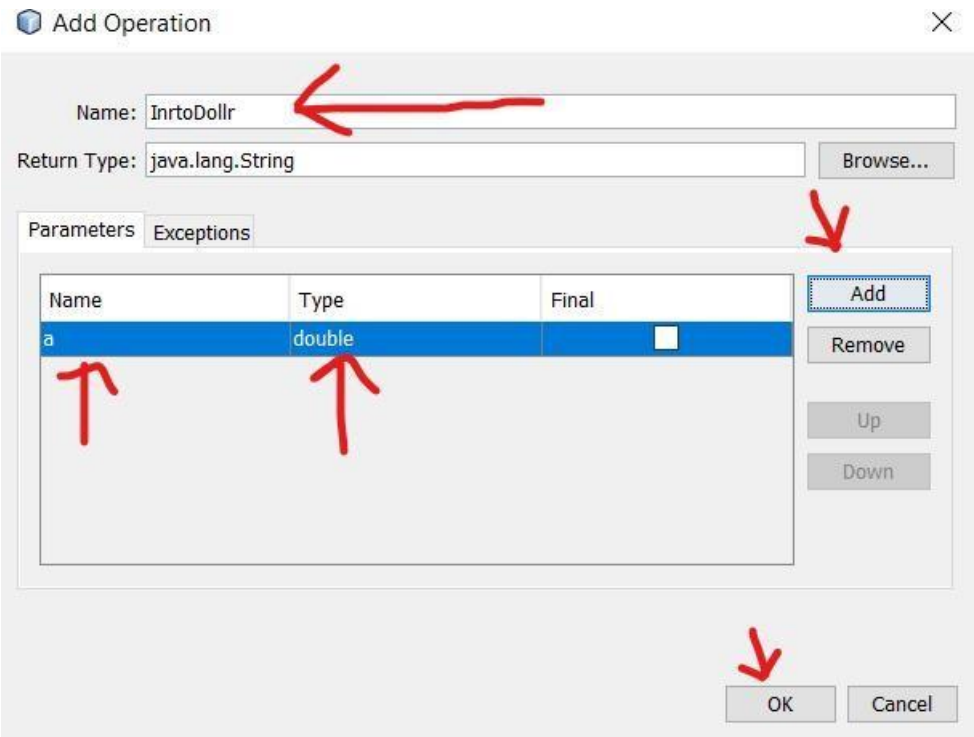


9] the following window would appear :

Just give name to the method or operation and click on add button to add parameters to the method as here we are converting dollar to rupees we should need only one parameter .



10] give the name to the variable select data type as double and click on ok as shown below:

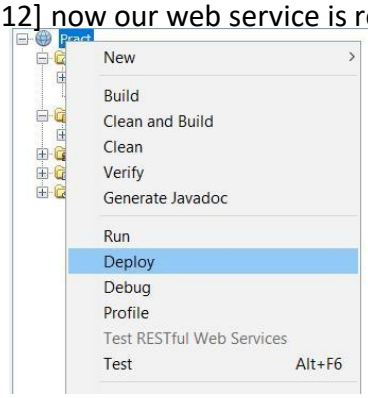


Fahrenheit to Celsius formula $(a - 32) \times 5/9$

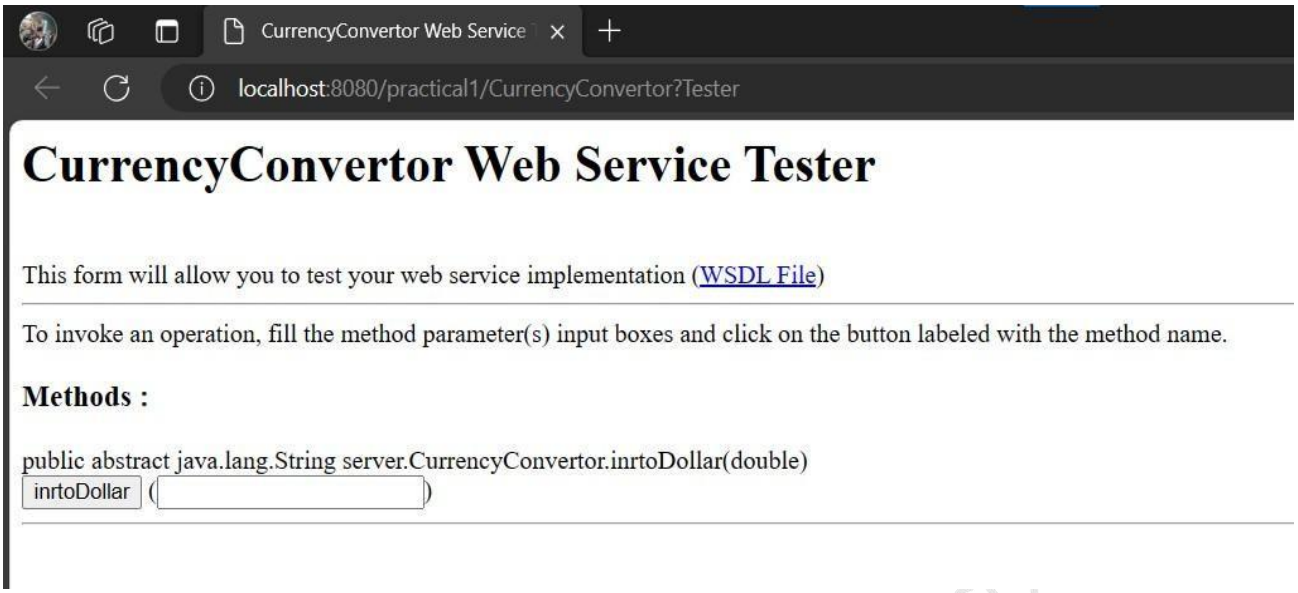
11] After clicking ok code will autogenerate , make changes In that code as mentioned below:

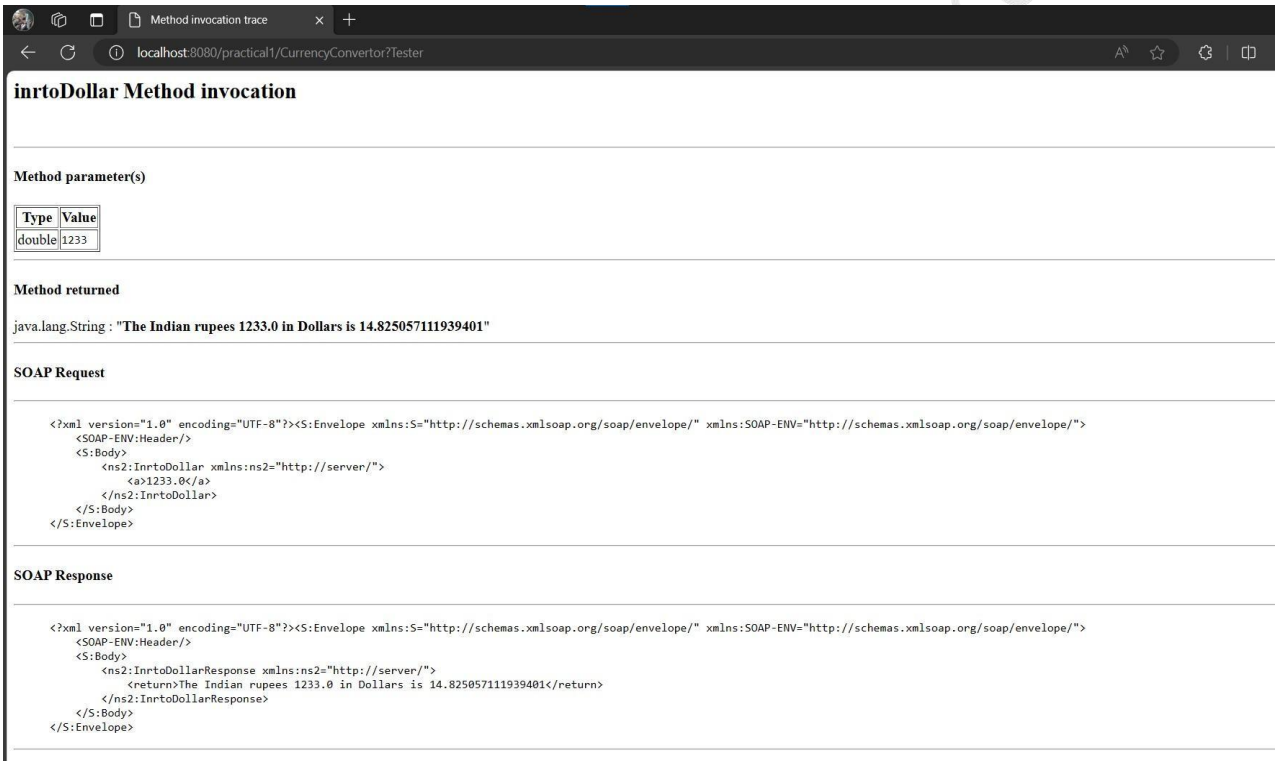
```
@WebMethod(operationName = "InrtoDollar")
public String InrtoDollar(@WebParam(name = "a") double a) {
    //TODO write your implementation code here:
    return "The Indian rupees "+a+" in Dollars is "+(a/83.17);
}
```

12] now our web service is ready now right click on project and click on deploy :



13] now right click on webservice and click on test web service you will get the following output:

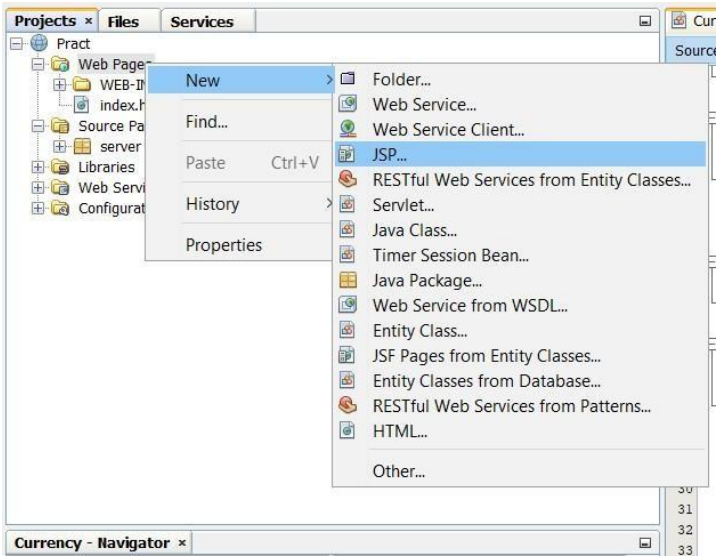




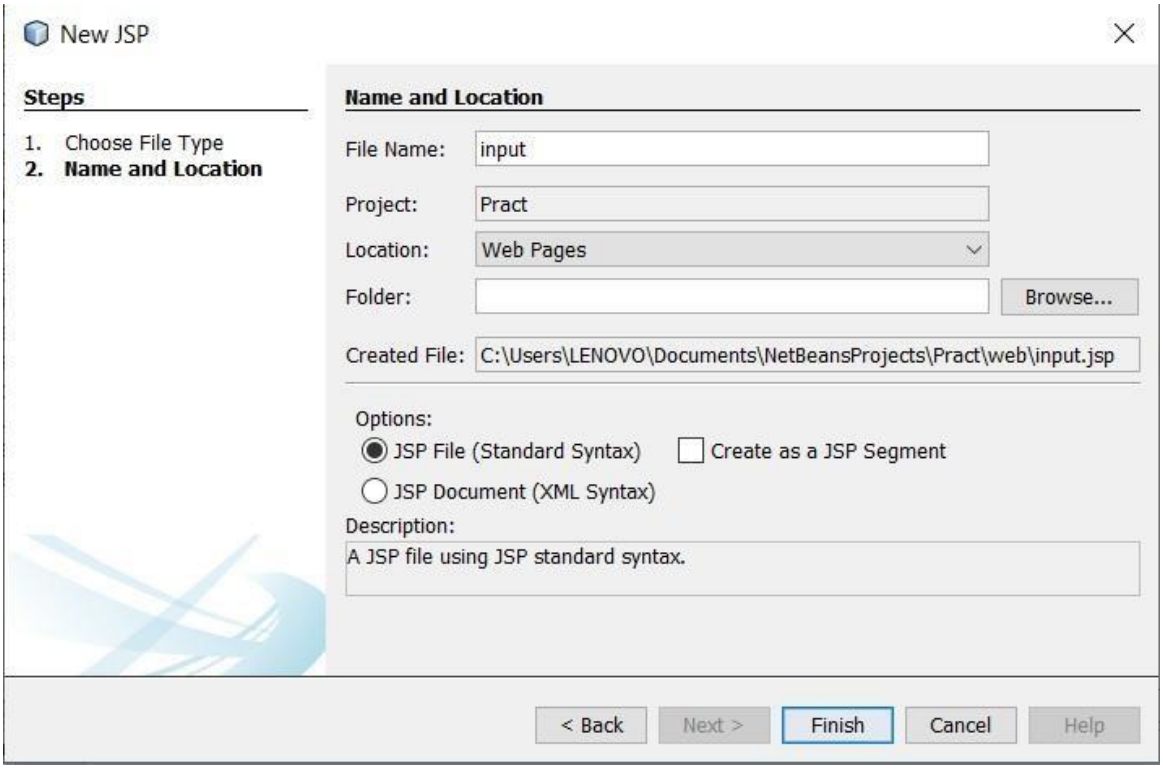
So this is how we created our web service and deployed it.

Creating a java Client using jsp

14] now our web service is successfully deployed. Right click on web pages and select new and select jsp as shown below:



15] give name and click on finish as shown below do it 2 times on for input and one foroutput:

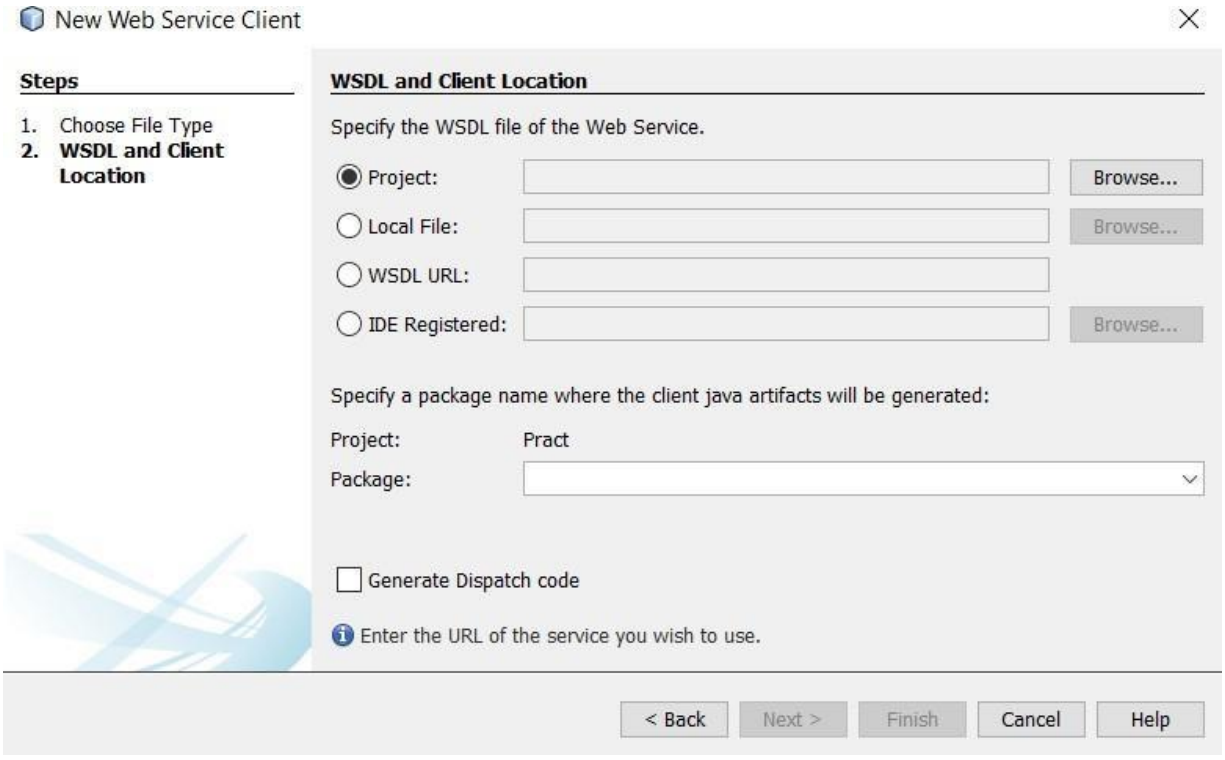


16] In input.jsp create a form for taking user input as shown below:

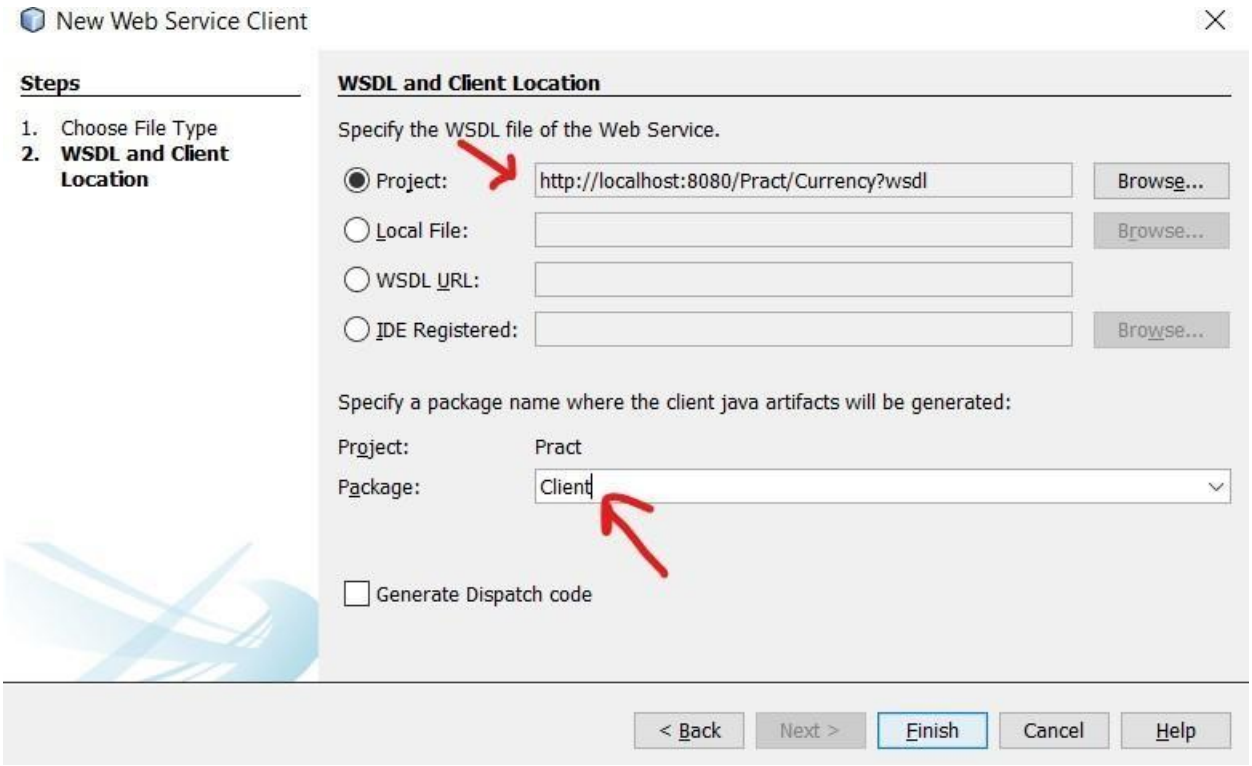
```
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>
        <form action="output.jsp">
            <pre>
                Enter the currency in rupees : <input type="text" name="t1">
                <input type="submit">       <input type="reset">
            </pre>
        </form>
    </body>
</html>
```

In this code set action = the jsp file where u want output and give name to textbox input.

17] now we have to create web service client, for same right click on project and select new then select web service client you will get the following screen:



At this step click on browse and select your project and click ok after clicking ok you will get wsdl url as shown below:



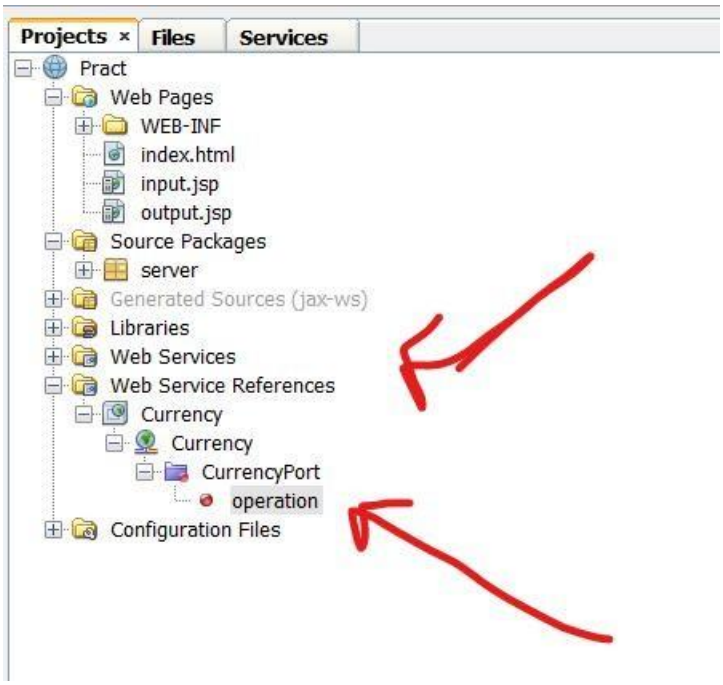
At this stage copy the url and paste it in notepad for future.

Give package name as “Client”.

Now click on finish.

18] now in project section you can se a new folder is been created named “web servicerefrence”.

double click to expand it you should get the following :



Hold the operation or your operation name and drag it into the output.jsp in body tag as shown : You will get the following auto generated code:

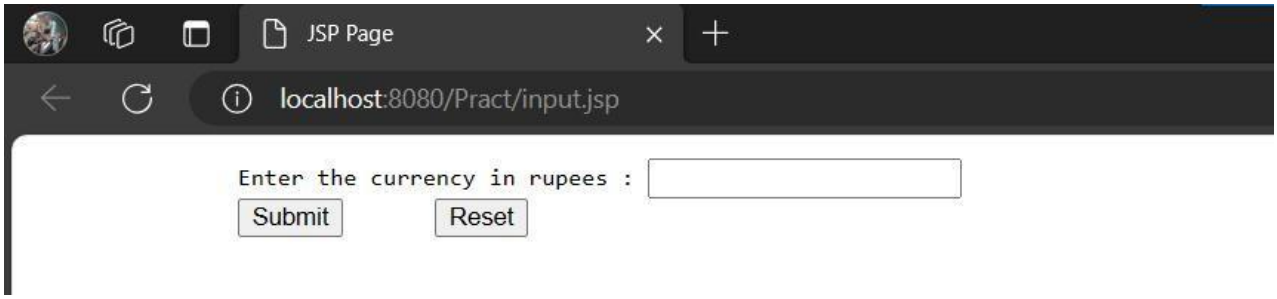
```
<body>
        <%-- start web service invocation --%><hr/>
<%
try {
    Client.Currency_Service service = new Client.Currency_Service();
    Client.Currency port = service.getCurrencyPort();
     // TODO initialize WS operation arguments here
    double a = 0.0d;
    // TODO process result here
    java.lang.String result = port.operation(a);
    out.println("Result = "+result);
} catch (Exception ex) {
    // TODO handle custom exceptions here
}
%>
<%-- end web service invocation --%><hr/>

</body>
```

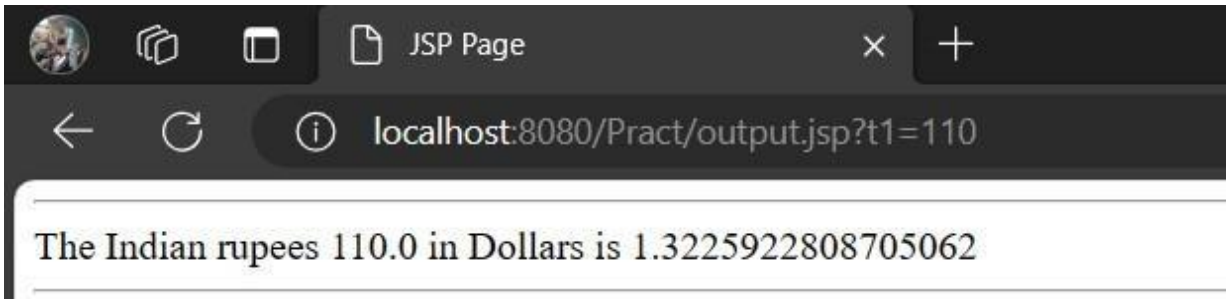
19] now make changes as shown below in the code:

```
        <%-- start web service invocation --%><hr/>
<%
try {
    Client.Currency_Service service = new Client.Currency_Service();
    Client.Currency port = service.getCurrencyPort();
     // TODO initialize WS operation arguments here
    double a = Double.parseDouble(request.getParameter("t1"));
    // TODO process result here
    java.lang.String result = port.operation(a);
    out.println(result);
} catch (Exception ex) {
    // TODO handle custom exceptions here
}
%>
<%-- end web service invocation --%><hr/>
```

20] now Deploy our project again and right click on input.jsp and click run file you will redirect to a browser page as shown :



Enter any numerical value in text box and click on submit you will get the following output:



Python client:

1] for consuming java web services in python we should repeat the above steps till step np.19 .

Note: for getting output the project or web service should be deployed .

After doing all of the steps write the following code in python idle:

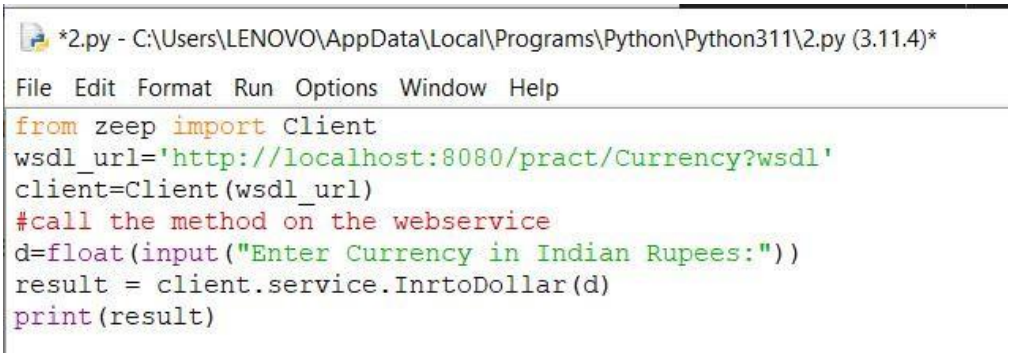
Code :

```
from zeep import Client
wsdl_url='http://localhost:8080/pract/Currency?wsdl'
client=Client(wsdl_url)
#call the method on the webservice
d=float(input("Enter Currency in Indian Rupees:"))
result = client.service.InrtoDollar(d)
print(result)
```

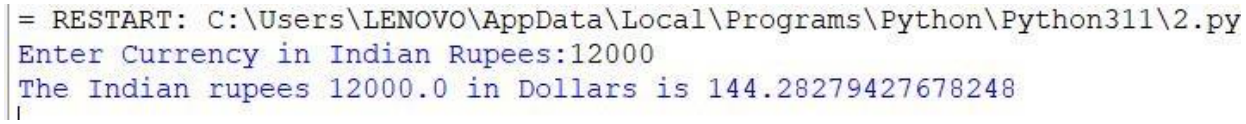
2] replace wsdl_url with the url which was copied at the time of web service client creation .

In step no.17 . and replace the InrtoDollarwith your method or operation name. and pass parameter to it

3] final code should look like following:

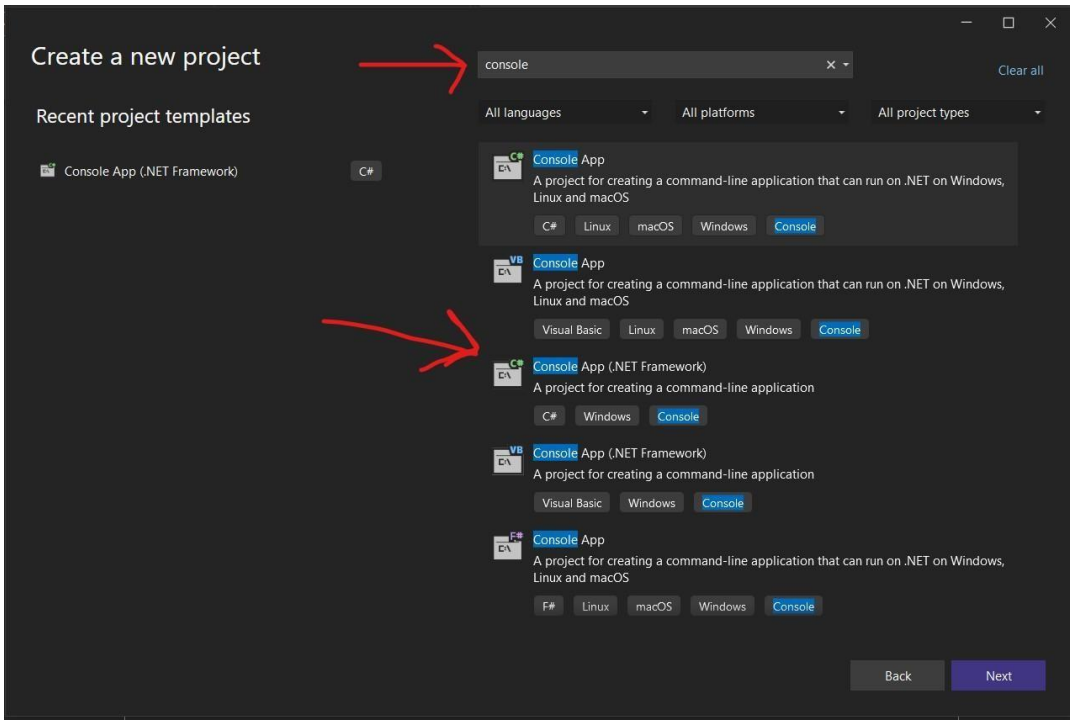


4] after running the above code you should get the following output:



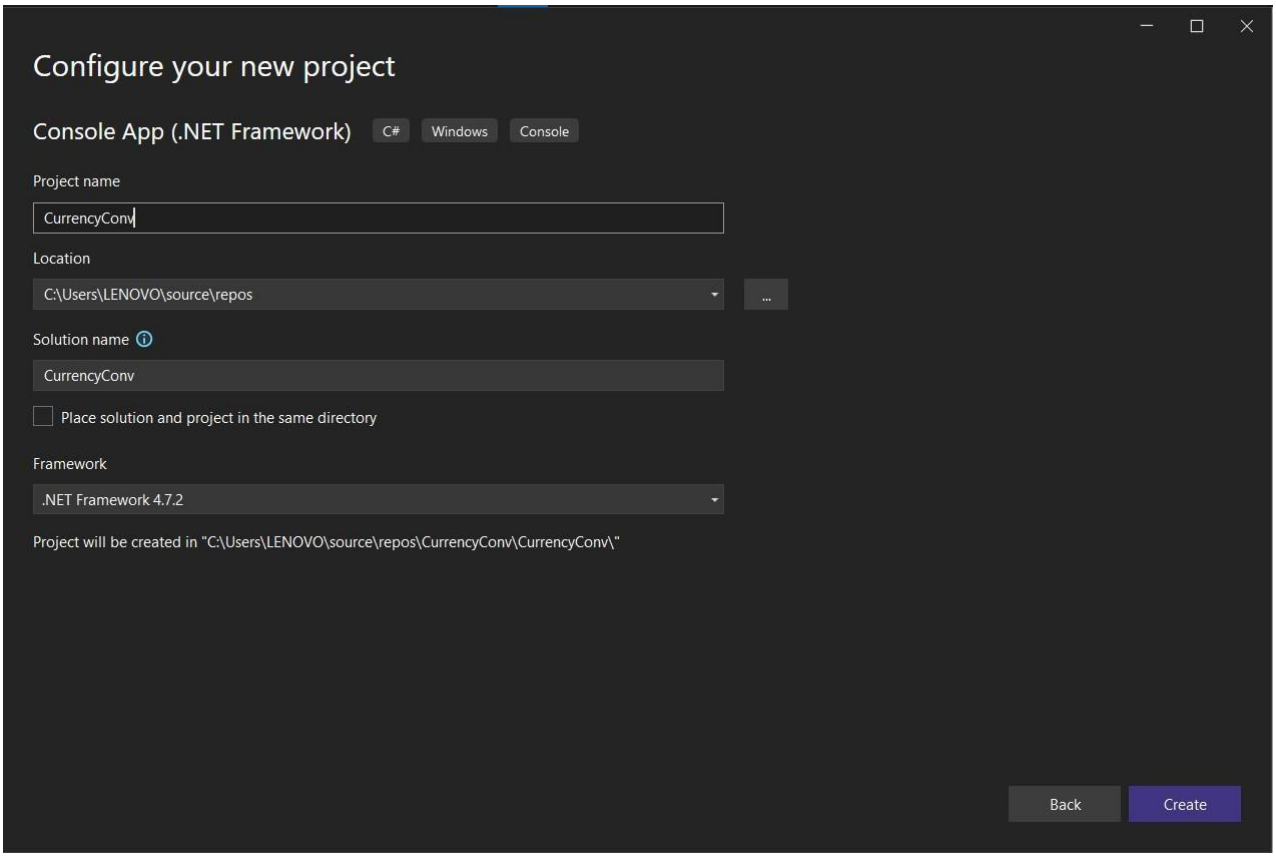
.Net Client

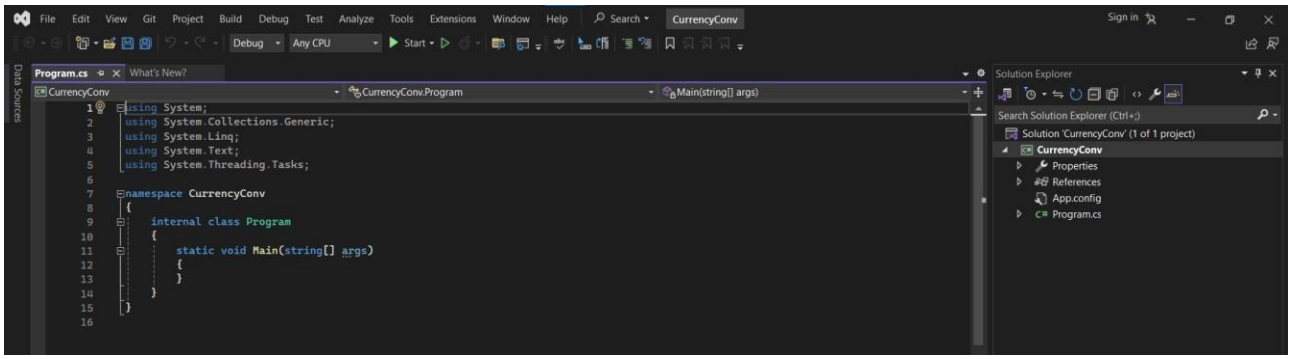
1] open visual studio and select create a new project the following window will appear:



Here in search bar type “console c#” and select console app for .Net frame work as shown in above image.

2] Give name to the project and press create button as shown in the image:

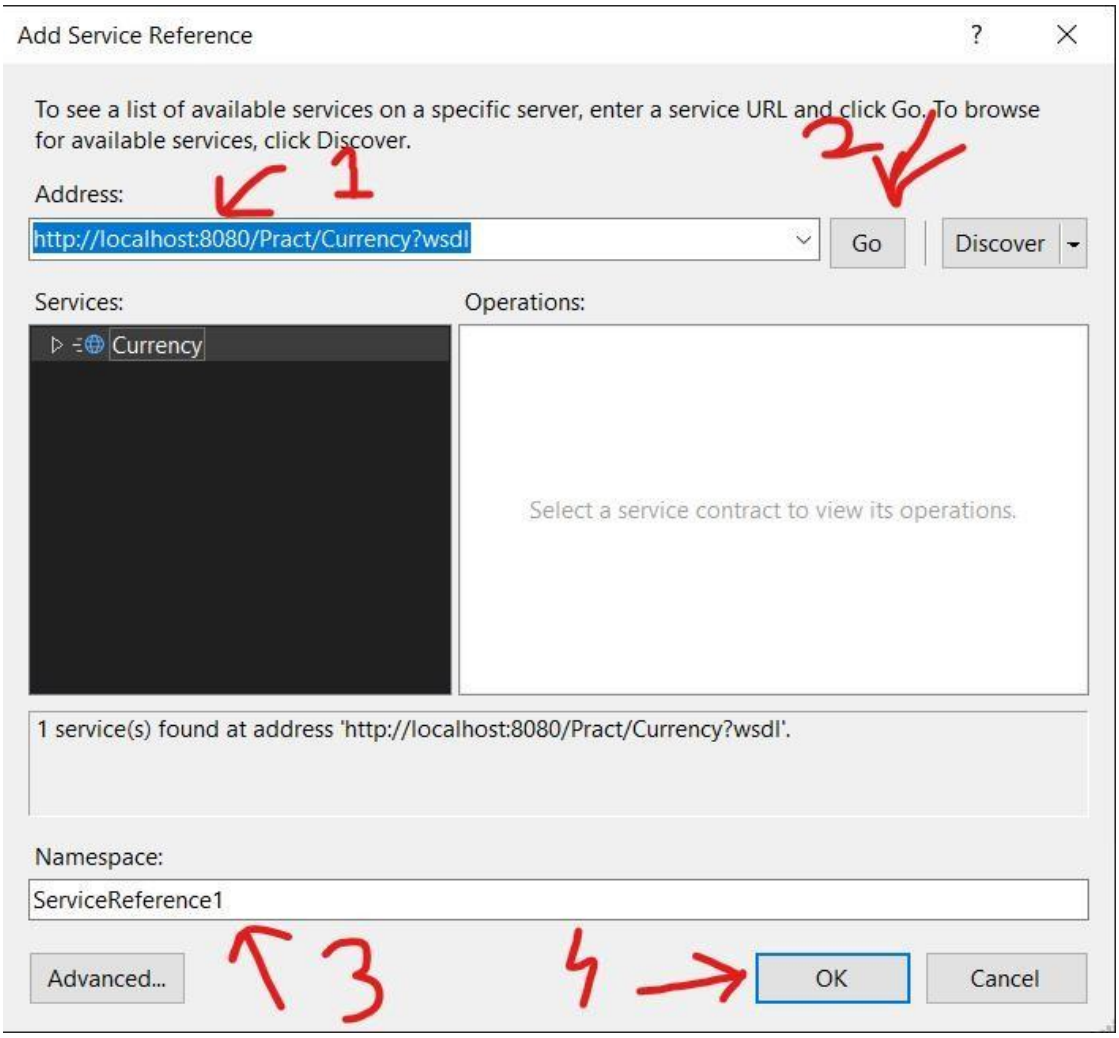




This window should appear after clicking create button now In right side there is solution Explorer.

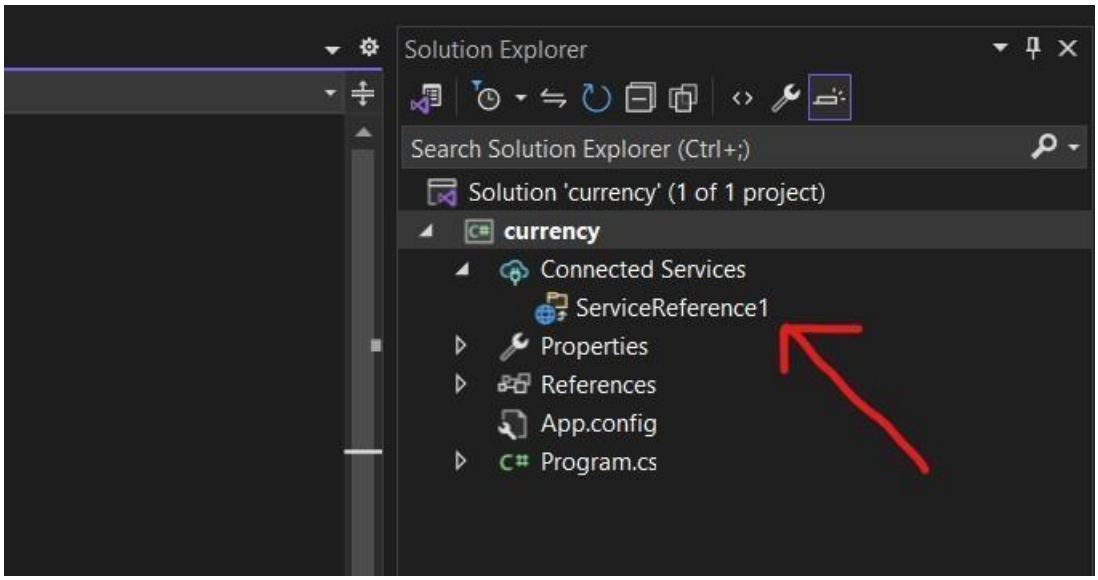
In that right click on your project name and click on add > Service Reference . You will get the following window :

Note: before executing the code make sure that the web service is deployed.

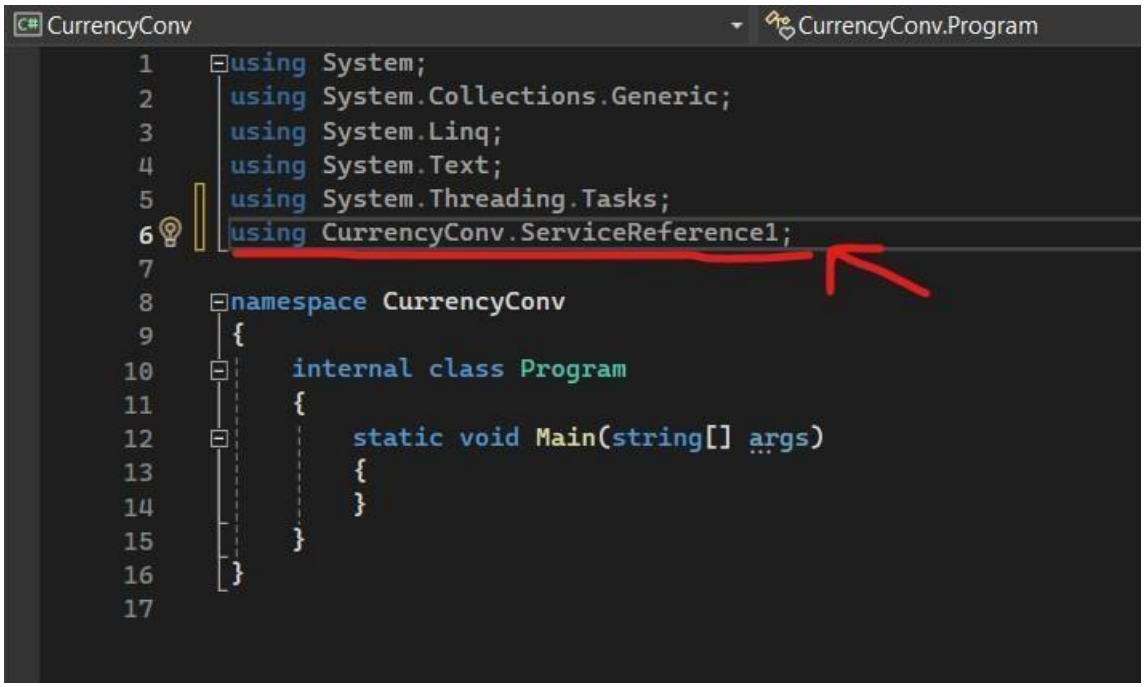


Here in address bar paste the wsdl url the same which is used I python code. And press go button . agter set the namespace In my case I am letting is as it is then press ok button.

In following image u can check in solution Explorer service refrence iscreated.:



3] now in code write the following line as shown in image:



Here type:

Using your_projectname . your_serviceReference_name ;

4] now write the code as shown in the following image:

Note:

Here CurrencyClient is my webservice class name and **operation() is my method name** as showyou can also check your own: